

## LECTURES 12-13

## Chapter 6 : Random vectors, joint distributions

*Key words:* Random vector, Borel  $\sigma$ -field of subsets of  $\mathbb{R}^n$ , joint distribution function, law of random vector or joint distribution, marginal distribution functions, conditional pmf, conditional pdf.

In many problems, one often encounter multiple random objects, for example: future price of two different stocks in a stock market. Since the price of a stock can affect the price of the second, it is not wise to analyze them separately. One should to analyze multiple random objects together, since behavior of one can affect others. To model such phenomenon, we need to introduce many random variables in a single platform (i.e., in a probability space).

First we will recall, some elementary facts about  $n$ -dimensional Euclidean space. Consider

$$\mathbb{R}^n = \overbrace{\mathbb{R} \times \cdots \times \mathbb{R}}^{n \text{ times}}$$

with the usual metric  $d(x, y) = \|x - y\|$ , where

$$\|x\| = \sqrt{\sum_{i=1}^n |x_i|^2}, \quad x = (x_1, \dots, x_n) \in \mathbb{R}^n.$$

A subset  $O$  of  $\mathbb{R}^n$  is said to be open if for each  $x \in O$ , there exists an  $\epsilon > 0$  such that

$$B(x, \epsilon) \subseteq O,$$

where

$$B(x, \epsilon) = \{y \in \mathbb{R}^n \mid \|x - y\| < \epsilon\}.$$

i.e., each point in an open set is an interior point. A useful fact: Any open set can be written as a countable union of open sets of the form  $\Pi_{i=1}^n(a_i, b_i)$ ,  $a_i < b_i$ , they are called open rectangles.

**Definition 6.1.** The  $\sigma$ -field generated by all open sets in  $\mathbb{R}^n$  is called the Borel  $\sigma$ -field of subsets of  $\mathbb{R}^n$  and is denoted by  $\mathcal{B}_{\mathbb{R}^n}$ .

Some Borel sets : Rectangles, triangles, lines, points all are Borel sets in  $\mathbb{R}^2$ .

**Theorem 0.1** *Let*

$$\mathcal{I}_n = \{(-\infty, x_1] \times \cdots \times (-\infty, x_n] \mid (x_1, \dots, x_n) \in \mathbb{R}^n\}.$$

*Then*

$$\sigma(\mathcal{I}_n) = \mathcal{B}_{\mathbb{R}^n}.$$

**Proof.** (Reading exercise) We prove for  $n = 2$ , for  $n \geq 3$ , it is similar. Note that

$$\mathcal{I}_2 \subseteq \mathcal{B}_{\mathbb{R}^2}.$$

Hence from the definition of  $\sigma(\mathcal{I}_2)$ , we have

$$\sigma(\mathcal{I}_2) \subseteq \mathcal{B}_{\mathbb{R}^2}.$$

Note that for  $(x_1, x_2) \in \mathbb{R}^2$ ,

$$(-\infty, x_1) \times (-\infty, x_2) = \bigcup_{m=1}^{\infty} \left[ \left( -\infty, x_1 - \frac{1}{m} \right] \times \left( -\infty, x_2 - \frac{1}{m} \right] \right] \in \sigma(\mathcal{I}_2).$$

For each  $x_1, x_2, y_1, y_2 \in \mathbb{R}$  such that  $x_1 < y_1, x_2 < y_2$  we have

$$\begin{aligned} (x_1, y_1) \times (x_2, y_2) &= (-\infty, y_1) \times (-\infty, y_2) \\ &\quad \setminus \left[ (-\infty, x_1] \times (-\infty, x_2] \cup (-\infty, x_1] \times (-\infty, y_2] \right. \\ &\quad \left. \cup (-\infty, y_1] \times (-\infty, x_2] \right]. \end{aligned}$$

Hence all bounded open rectangles are in  $\sigma(\mathcal{I}_2)$ . Since any open set in  $\mathbb{R}^2$  can be rewritten as a countable union of bounded open rectangles, all open sets are in  $\sigma(\mathcal{I}_2)$ . Therefore from the definition of  $\mathcal{B}_{\mathbb{R}^2}$ , we get

$$\mathcal{B}_{\mathbb{R}^2} \subseteq \sigma(\mathcal{I}_2).$$

This completes the proof. (It is advised that student try to write down the proof for  $n = 3$ )

**Definition 6.2.** Let  $\Omega$  be a non empty set (sample space) and  $\mathcal{F}$  be a  $\sigma$ -field of subsets of  $\Omega$ . A map  $X : \Omega \rightarrow \mathbb{R}^n$ , is called a random vector/  $\mathbb{R}^n$ -valued random variable (with respect to  $\mathcal{F}$ ) if

$$X^{-1}(B) \in \mathcal{F} \text{ for all } B \in \mathcal{B}_{\mathbb{R}^n}.$$

Note that  $X = (X_1, \dots, X_n)$ , where  $X_i : \Omega \rightarrow \mathbb{R}$  denote the  $i$ th component of  $X$ .

**Theorem 0.2**  $X$  is a random vector with respect to  $\mathcal{F}$  iff

$$\{X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n\} \in \mathcal{F} \text{ for all } x_1, x_2, \dots, x_n \in \mathbb{R}.$$

**Proof:** if part follows from the definition. We prove the only if part.

Suppose  $X : \Omega \rightarrow \mathbb{R}^n$  satisfies

$$\{X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n\} \in \mathcal{F} \text{ for all } x_1, x_2, \dots, x_n \in \mathbb{R}.$$

Let

$$\mathcal{D} = \{B \in \mathcal{B}_{\mathbb{R}^n} | X^{-1}(B) \in \mathcal{F}\}.$$

Clearly

$$\mathcal{I}_n \subseteq \mathcal{D}.$$

Now it is easy to check that  $\mathcal{D}$  is a  $\sigma$ -field (exercise). Hence

$$\sigma(\mathcal{I}_n) \subseteq \mathcal{D} \subseteq \mathcal{B}_{\mathbb{R}^n}.$$

Therefore using Theorem 1.1, we have  $\mathcal{D} = \mathcal{B}_{\mathbb{R}^n}$ . i.e. for each  $B \in \mathcal{B}_{\mathbb{R}^n}$ ,  $X^{-1}(B) \in \mathcal{F}$ . This completes the proof.

The  $\sigma$ -field generated by  $X$  is defined by

$$\sigma(X) = \{X^{-1}(B) | B \in \mathcal{B}_{\mathbb{R}^n}\}.$$

(exercise)

$$\sigma(X) = \sigma(\sigma(X_1) \cup \dots \cup \sigma(X_n)).$$

Now we introduce Borel functions in the current context.

**Definition 0.1** A function  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is said to be a Borel function if

$$f^{-1}(B) \in \mathcal{B}_{\mathbb{R}^n}, \forall B \in \mathcal{B}_{\mathbb{R}^m}.$$

**Lemma 0.1** *If  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is continuous, then it is a Borel function.*

**Proof:** The main ingredient of the proof is the following fact.  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  is continuous iff  $f^{-1}(O)$  is open for each open set  $O \subseteq \mathbb{R}^m$ .

Using this, the proof is standard, so is left as an exercise.  $\square$

**Example 0.1** 1.  $f_1(x, y) = xy$ ,  $f_2(x, y) = x + y$  are Borel functions, since they are continuous functions.

2. Define  $\pi_i : \mathbb{R}^n \rightarrow \mathbb{R}$  as follows: for each  $i = 1, 2, \dots, n$ ,

$$\pi_i(x_1, x_2, \dots, x_n) = x_i, (x_1, x_2, \dots, x_n) \in \mathbb{R}^n.$$

Then  $\pi_i$  is the  $i$ th coordinate projection map and is continuous and hence is a Borel function.

**Exercise 6.1** Let  $B$  is a Borel set in  $\mathbb{R}$ , then show that  $B \times \mathbb{R}$  is a Borel set in  $\mathbb{R}^2$ .

Method 1: Use the following argument. Collect all Borel set  $B$  satisfying " $B \times \mathbb{R}$  is Borel set in  $\mathbb{R}^2$ ". Then show that it is a  $\sigma$ -field. Also show that this collection contains all open sets in  $\mathbb{R}$ . Now show that (exercise) the collection is indeed all Borel sets in  $\mathbb{R}^2$ .

Method 2: Note that (exercise)  $\pi_1^{-1}(B) = B \times \mathbb{R}$ . Since  $\pi_1$  is a Borel function, it follows that  $B \times \mathbb{R} = \pi_1^{-1}(B) \in \mathcal{B}_{\mathbb{R}^2}$  for  $B \in \mathcal{B}_{\mathbb{R}}$ .

**Lemma 0.2** *Let  $X$  be an  $\mathbb{R}^n$ -valued random variable and  $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be a Borel function. Then  $\varphi(X)$  is an  $\mathbb{R}^m$ -valued random variable.*

**Proof.** For  $B \in \mathcal{B}_{\mathbb{R}^m}$ ,

$$\varphi(X)^{-1}(B) = X^{-1}(\varphi^{-1}(B)).$$

Since  $\varphi^{-1}(B) \in \mathcal{B}_{\mathbb{R}^m}$  and since  $X$  is an  $\mathbb{R}^n$ -valued random variable, we get  $\varphi(X)^{-1}(B) \in \mathcal{F}$ . Hence  $\varphi(X)$  is an  $\mathbb{R}^m$ -valued random variable.  $\square$

**Theorem 0.3**  *$X : \Omega \rightarrow \mathbb{R}^n$  is a random vector iff  $X_i, i = 1, 2, \dots, n$  are random variables where  $X_i$  denote the  $i^{\text{th}}$  component of  $X$ .*

**Proof:** Let  $X$  be a random vector. Since  $X_i = \pi_i(X)$ ,  $i = 1, 2, \dots, n$ , using Lemma 1.2, it follows that  $X_i$  is a random variable.

Suppose  $X_1, X_2, \dots, X_n$  are random variables.

For  $x_1, x_2, \dots, x_n \in \mathbb{R}$ ,

$$\{X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n\} = \{X_1 \leq x_1\} \cap \{X_2 \leq x_2\} \cap \dots \cap \{X_n \leq x_n\} \in \mathcal{F}.$$

This completes the proof.  $\square$

**Theorem 0.4** Let  $X = (X_1, X_2, \dots, X_n)$  be a random vector defined on a probability space  $(\Omega, \mathcal{F}, P)$ . On  $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$  define  $\mu$  as follows

$$\mu(B) = P(X \in B).$$

Then  $\mu$  is a probability measure on  $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$ .

**Proof.** Since  $\{X \in \mathbb{R}^n\} = \Omega$ , we have

$$\mu(\mathbb{R}^n) = 1.$$

Let  $B_1, B_2, \dots$  be pair wise disjoint elements from  $\mathcal{B}_{\mathbb{R}^n}$ . Then  $X^{-1}(B_1), X^{-1}(B_2), \dots$  are pair wise disjoint and are in  $\mathcal{F}$ . Hence

$$\mu\left(\bigcup_{m=1}^{\infty} B_m\right) = P\left(\bigcup_{m=1}^{\infty} X^{-1}(B_m)\right) = \sum_{m=1}^{\infty} P(X \in B_m) = \sum_{m=1}^{\infty} \mu(B_m).$$

This completes the proof.  $\square$

**Definition 6.3.** The probability measure  $\mu$  given in Theorem 1.4 is called the distribution (or Law) of the random vector  $X$  and is denoted by  $\mu_X$ .

**Definition 6.4.** (joint distribution function)

Let  $X = (X_1, X_2, \dots, X_n)$  be a random vector. Then the function  $F : \mathbb{R}^n \rightarrow \mathbb{R}$  given by

$$F(x_1, x_2, \dots, x_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

is called the distribution function of  $X$  (in otherwords, joint distribution function of the random variables  $X_1, X_2, \dots, X_n$ ).

Now onwards **we set**  $n = 2$ . Let  $x_1 \wedge x_2$  denote the minimum and  $x_1 \vee x_2$  denote the maximum of the real numbers  $x_1, x_2$ .

**Theorem 0.5** *Let  $F$  be the joint distribution function of a random vector  $X$ . Then  $F$  satisfies the following.*

(i) (a)

$$\lim_{x_1 \wedge x_2 \rightarrow -\infty} F(x_1, x_2) = 0,$$

(b)

$$\lim_{x_1 \wedge x_2 \rightarrow \infty} F(x_1, x_2) = 1$$

(ii)  $F$  is right continuous, i.e.

$$\lim_{(y_1, y_2) \searrow (x_1, x_2)} F(y_1, y_2) = F(x_1, x_2) \text{ for all } x_1, x_2 \in \mathbb{R},$$

here  $(y_1, y_2) \searrow (x_1, x_2)$  means  $x_i \leq y_i, i = 1, 2$  and  $(y_1, y_2) \rightarrow (x_1, x_2)$ .

(iii)  $F$  is with non negative increments, i.e. given  $a = (a_1, a_2), b = (b_1, b_2)$  with  $a_1 < b_1, a_2 < b_2$ ,

$$F(\mathbf{a}, \mathbf{b}] \geq 0,$$

where  $(\mathbf{a}, \mathbf{b}] := (a_1, b_1] \times (a_2, b_2]$  and

$$F(\mathbf{a}, \mathbf{b}] := F(b_1, b_2) - F(b_1, a_2) - F(a_1, b_2) + F(a_1, a_2).$$

The proof of the above theorem is an easy exercise to the student. For example to prove (iii), observe that

$$F(\mathbf{a}, \mathbf{b}] = P(X \in (\mathbf{a}, \mathbf{b}]).$$

**Remark 0.1** *As in the random variable case (i.e. with  $n = 1$ ), the above properties (i) to (iii) becomes characterizing properties for distribution function, i.e. if  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$  satisfies (i) to (iii), then there exists a random vector  $X$  such that  $F$  is the distribution function of  $X$ .*

Given a random vector  $X = (X_1, X_2)$ , the distribution function of  $X_1$  denoted by  $F_{X_1}$  is called the marginal distribution of  $X_1$ . Similarly the marginal distribution function  $F_{X_2}$  of  $X_2$  is defined. Given the joint distribution function  $F$  of  $X$ , one can recover the corresponding marginal distributions as follows.

$$F_{X_1}(x_1) = P(X_1 \leq x_1) = P(X_1 \leq x_1, X_2 \in \mathbb{R}) = \lim_{x_2 \rightarrow \infty} F(x_1, x_2).$$

Similarly

$$F_{X_2}(x_2) = \lim_{x_1 \rightarrow \infty} F(x_1, x_2).$$

Given the marginal distribution functions of  $X_1$  and  $X_2$ , in general it is impossible to construct the joint distribution function. Note that marginal distribution functions doesn't contain information about the dependence of  $X_1$  over  $X_2$  and vice versa. One can characterize the independence of  $X_1$  and  $X_2$  in terms of its joint and marginal distribution functions as in the following theorem. The proof is beyond the scope of this course.

**Theorem 0.6** *Let  $X = (X_1, X_2)$  be a random vector with distribution function  $F$ . Then  $X_1$  and  $X_2$  are independent iff*

$$F(x_1, x_2) = F_{X_1}(x_1) F_{X_2}(x_2), \quad x_1, x_2 \in \mathbb{R}.$$

Let us recall some notations.

$$\begin{aligned} F(a-, b-) &:= \lim_{(x,y) \uparrow (a,b)} F(x, y) \\ F(a-, b) &:= \lim_{x \uparrow a} F(x, b) \\ F(a, b-) &:= \lim_{y \uparrow b} F(a, y), \end{aligned}$$

where  $(x, y) \nearrow (a, b)$  geometrically may be viewed as  $(-\infty, x] \times (-\infty, y] \uparrow (-\infty, a) \times (-\infty, b)$ . One can easily see that

$$\begin{aligned} F(a-, b-) &= P(X < a, Y < b), \quad F(a-, b) = P(X < a, Y \leq b), \\ F(a, b-) &= P(X \leq a, Y < b). \end{aligned}$$

**Lemma 0.3** *Let  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a distribution function and  $(X, Y)$  be a corresponding random vector and let  $x, y \in \mathbb{R}$ .*

(i)

$$P(X = x, Y = y) = F(x, y) - F(x-, y) - F(x, y-) + F(x-, y-).$$

(ii)  $F$  is continuous at  $(x, y)$  iff  $F(x, y) = F(x-, y-)$ .

**Proof:** (Reading Exercise)

Proof of (i)

Note that

$$\bigcap_{n=1}^{\infty} \left\{ x - \frac{1}{n} < X \leq x, y - \frac{1}{n} < Y \leq y \right\} = \{X = x, Y = y\}.$$

Hence

$$\begin{aligned} P(X = x, Y = y) &= \lim_{n \rightarrow \infty} P\left(x - \frac{1}{n} < X \leq x, y - \frac{1}{n} < Y \leq y\right) \\ &= \lim_{n \rightarrow \infty} \left(F(x, y) - F\left(x - \frac{1}{n}, y\right) - F\left(x, y - \frac{1}{n}\right) + F\left(x - \frac{1}{n}, y - \frac{1}{n}\right)\right) \\ &= F(x, y) - F(x-, y) - F(x, y-) + F(x-, y-). \end{aligned}$$

Proof of (ii) Follows from

$$F(x-, y-) \leq F(x-, y) \leq F(x, y), \quad F(x-, y-) \leq F(x, y-) \leq F(x, y).$$

**Definition 6.5.** (i) A distribution function  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$  is said to be of discrete type if there exists a countable set  $\{(x_i, y_i) | i \in I\}$  such that

$$\sum_{i \in I} (F(x_i, y_i) - F(x_i-, y_i) - F(x_i, y_i-) + F(x_i-, y_i-)) = 1.$$

( in terms of the corresponding random vector,  $(X, Y)$  is discrete if

$$\sum_{i \in I} P(X = x_i, Y = y_i) = 1. )$$

(ii) (pmf of discrete random vector) Define the function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  by

$$f(x, y) = F(x, y) - F(x-, y) - F(x, y-) + F(x-, y-), \quad x, y \in \mathbb{R}.$$

Then  $f$  is called pmf of  $F$ .

**Remark 0.2**  $(X, Y)$  is discrete iff  $X$  and  $Y$  are discrete random variables.

Let  $X, Y$  be discrete random variables with distribution functions  $F_1, F_2$ . Let  $D_i$  denote respectively the set of discontinuities of  $F_i, i = 1, 2$ . Then

$$\begin{aligned} \sum_{x \in D_1, y \in D_2} P(X = x, Y = y) &= \sum_{x \in D_1} P(X = x, Y \in D_2) \\ &= \sum_{x \in D_1} P(X = x) = 1. \end{aligned}$$

Hence  $(X, Y)$  is discrete. Converse follows by a similar argument. (exercise)

**Example 0.2** Consider the experiment of throwing a die twice independently. Let  $X_1$  denote the number of 1's and  $X_2$  denote the number of 3's. We will write down the joint pmf of  $X_1$  and  $X_2$ .

| $X_1/X_2$ | 0              | 1              | 2              |
|-----------|----------------|----------------|----------------|
| 0         | $\frac{4}{9}$  | $\frac{2}{9}$  | $\frac{1}{36}$ |
| 1         | $\frac{2}{9}$  | $\frac{1}{18}$ | 0              |
| 2         | $\frac{1}{36}$ | 0              | 0              |

I will illustrate how the pmf is calculated.  $D_i$  denote the face shown by the die in the  $i$ th throw.

$$\begin{aligned}
 f(0,0) &= P(X_1 = 0, X_2 = 0) \\
 &= P(D_1 \neq 1, 3 \& D_2 \neq 1, 3) \\
 &= \frac{4}{6} \times \frac{4}{6} = \frac{4}{9}.
 \end{aligned}$$

The above is an example of a lattice distribution, i.e. 'probability masses' only at the 'lattice' points  $(i, j)$ ;  $i, j \in \mathbb{Z}$ .

**Example 0.3** Consider the function  $F : \mathbb{R}^2 \rightarrow \mathbb{R}$  given by

$$F(x, y) = \begin{cases} 0 & \text{if } x < 0 \text{ or } y < 0 \\ \frac{1}{3} \min\{x, 1\} & \text{if } (x, y) \in [0, \infty) \times [0, 1) \\ \min\{x, 1\} & \text{if } (x, y) \in [0, \infty) \times [1, \infty). \end{cases}$$

$$F_1(x) := \lim_{y \rightarrow \infty} F(x, y) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1 \end{cases}$$

and

$$F_2(y) := \lim_{x \rightarrow \infty} F(x, y) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{3} & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases}$$

Then  $F_1$  is the distribution function of Uniform  $(0, 1)$  random variable, say  $X$  and  $F_2$  is the distribution function the Bernoulli  $(\frac{2}{3})$  random variable, say  $Y$ .

Also observe that  $F(x, y) = F_1(x)F_2(y)$ ,  $x, y \in \mathbb{R}$ . Hence  $F$  is the distribution of  $(X, Y)$  where  $X$  and  $Y$  are independent.

It is easy to verify that  $F$  is continuous on  $B_1 \cup B_2 \cup B_3 \cup B_4 \cup B_5$  where

$$B_1 = \{(x, y) | x < 0 \text{ or } y < 0\}, B_2 = (0, 1) \times (0, 1),$$

$$B_3 = (0, 1) \times (1, \infty), B_4 = (1, \infty) \times (0, 1), B_5 = (1, \infty) \times (1, \infty).$$

Consider the boundary sets

$$C_1 = \{(0, y) | 0 < y < 1\}, C_2 = \{(x, 0) | 0 < x < 1\},$$

$$C_3 = \{(1, y) | 0 < y < 1\}, C_4 = \{(x, 1) | 0 < x < 1\},$$

$$D_1 = \{(0, y) | 1 < y < \infty\}, D_2 = \{(1, y) | 1 < y < \infty\},$$

$$D_3 = \{(x, 0) | 1 < x < \infty\}, D_4 = \{(x, 1) | 1 < x < \infty\},$$

and the corner points  $(0, 0)$ ,  $(1, 0)$ ,  $(0, 1)$  and  $(1, 1)$ .

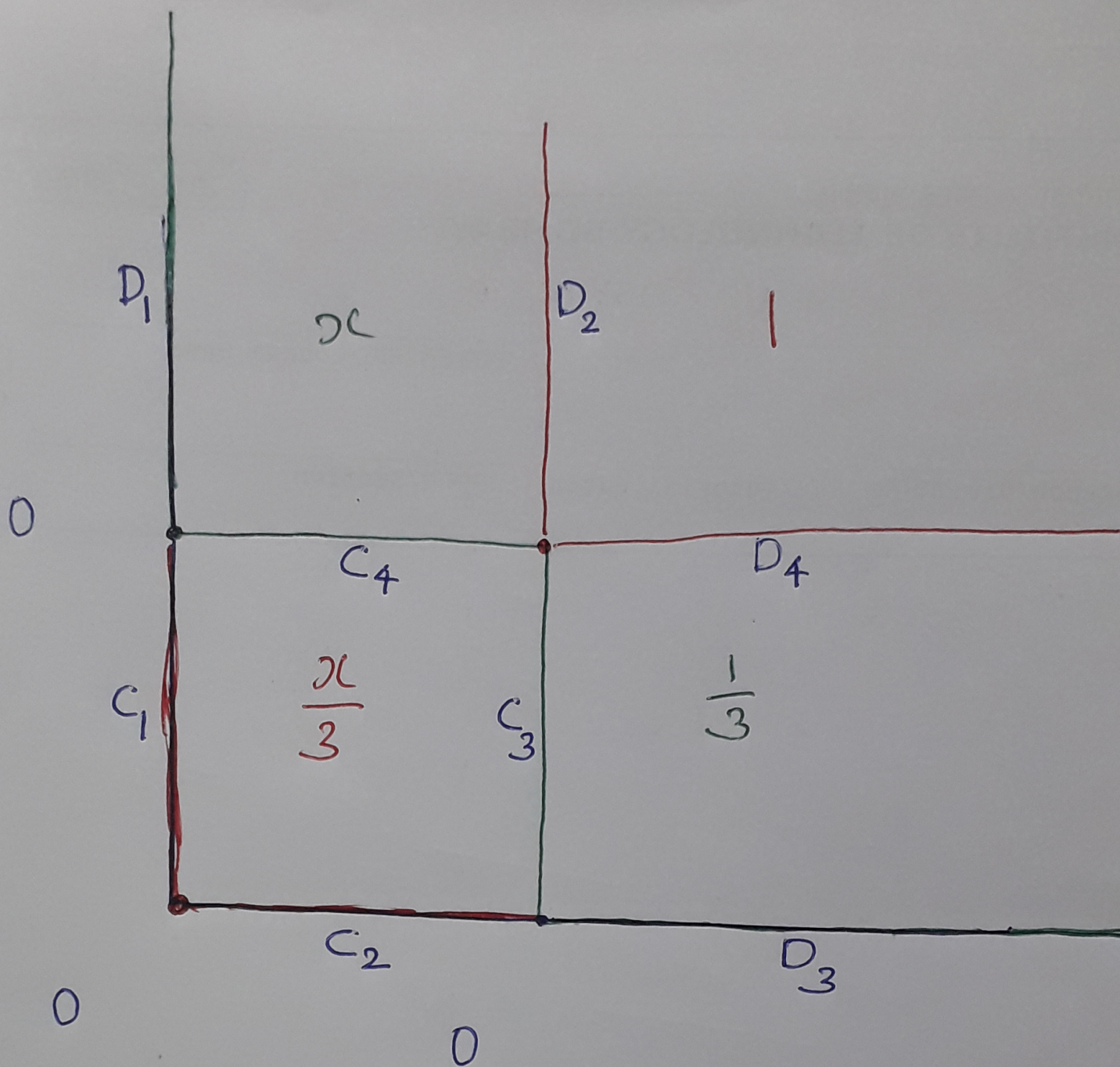
For  $(x, 0) \in C_2$ ,  $F(x-, 0-) = 0 \neq \frac{x}{3} = F(x, 0)$ . Hence  $F$  is discontinuous on  $C_2 \setminus \{0\}$ . Similarly, for  $(x, 1) \in C_3$ ,  $F(x-, 1-) = \frac{1}{3} = F(x, 1)$ . So  $F$  is continuous on  $C_3$ . For  $(0, y) \in C_1$ ,  $F(0-, y-) = 0 = F(0, y)$ , hence  $F$  is continuous on  $C_1$ .

For  $(x, 1) \in C_4$ ,  $F(x-, 1-) = \frac{x}{3} \neq x = F(x, 1)$ . Hence  $F$  is not continuous on  $C_4$ . Similarly  $F$  is not continuous on  $D_3$  and  $D_4$ .

Left limit at the corner points are  $F(0-, 0-) = 0 = F(0, 0)$ ,  $F(1-, 0-) = 0 \neq \frac{1}{3} = F(1, 0)$ ,  $F(0-, 1-) = 0 = F(0, 1)$ ,  $F(1-, 1-) = \frac{1}{3} \neq 1 = F(1, 1)$ .

Hence  $F$  is not left continuous at  $(1, 0)$  and  $(1, 1)$ .

Unit



**Example 0.4**

$$G(x, y) = \begin{cases} 0 & \text{if } x < 0 \text{ or } y < 0 \\ \frac{x}{4} & \text{if } (x, y) \in [0, 1) \times [0, 1) \\ x & \text{if } (x, y) \in [0, 1) \times [1, \infty) \\ \frac{1}{3} & \text{if } (x, y) \in [1, \infty) \times [0, 1) \\ 1 & \text{if } (x, y) \in [1, \infty) \times [1, \infty). \end{cases}$$

Verify that  $G$  is a distribution function by showing the properties (i), (ii) and (iii).

Let  $F_1, F_2$  be respectively be the distribution function of uniform  $(0, 1)$  and Bernoulli  $(\frac{2}{3})$  random variables. Then

$$\lim_{y \rightarrow \infty} G(x, y) = F_1(x), \quad \lim_{x \rightarrow \infty} G(x, y) = F_2(y).$$

(exercise)

**Example 0.5** Let  $X$  be uniform  $(0, 1)$  and  $F_2 : \mathbb{R} \rightarrow \mathbb{R}$  be the distribution function

$$F_2(y) := \lim_{x \rightarrow \infty} F(x, y) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{3} & \text{if } 0 \leq x < 1 \\ 1 & \text{if } x \geq 1. \end{cases}$$

Define  $Y = F_2^{-1}(X)$ , where

$$F_2^{-1}(\alpha) = \inf\{y \mid F_2(y) \geq \alpha\}.$$

Then marginal distribution of  $F_2^{-1}(X)$  is  $F_2$  (exercise). Hence we have another joint distribution function for uniform  $(0, 1)$  and Bernoulli  $(\frac{2}{3})$ .

**Definition 6.6.** (joint pdf of continuous random vector) Let  $(X, Y)$  be a continuous random vector (i.e., distribution function  $F$  is continuous). If there exists a function  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  such that

$$F(x, y) = \int_{-\infty}^y \int_{-\infty}^x f(t, s) dt ds, \quad \forall x, y \in \mathbb{R}$$

then  $f$  is called the pdf of  $(X, Y)$ , in otherwords, the joint pdf of the random variables  $X$  and  $Y$ .

**Theorem 0.7** Let  $(X, Y)$  be a continuous random vector with pdf  $f$ . Then

$$P\{(X, Y) \in B\} = \iint_B f(x, y) dx dy, \quad B \in \mathcal{B}_{\mathbb{R}^2}.$$

**Proof.** (Proof is not part of the syllabus and but the proof is given as a reading exercise which is optional )

Note that L.H.S of the equality corresponds to the law of  $(X, Y)$ .

Let  $\mathcal{F}_0$  denote the set of all finite union of rectangles in  $\mathbb{R}^2$ . Then  $\mathcal{F}_0$  is a field (exercise for the student).

Set

$$\begin{aligned} \mu_1(B') &= \iint_{B'} f(x, y) dx dy, \quad B' \in \mathcal{B}_{\mathbb{R}^2} \quad \text{and} \\ \mu_2(B') &= P\{(X, Y) \in B'\}, \quad B' \in \mathcal{B}_{\mathbb{R}^2} \end{aligned}$$

Then  $\mu_1, \mu_2$  are probability measures on  $\mathcal{B}_{\mathbb{R}^2}$  and  $\mu_1 = \mu_2$  on  $\mathcal{F}_0$ . Hence, using Kolmogorov's extension theorem, we have

$$\mu_1 = \mu_2 \text{ on } \mathcal{B}_{\mathbb{R}^2}.$$

i.e.,

$$P\{(X, Y) \in B\} = \iint_B f(x, y) dx dy, \quad B \in \mathcal{B}_{\mathbb{R}^2}.$$

**Remark 0.3** The integral in Theorem 1.7 is in general not understood in the Riemann integral sense but for all our computations, the integral will be in the Riemann sense. Note mostly will be considering Riemann integrable functions in domains which are either rectangles or of the form

$$B = \{(x, y) | a \leq x \leq b, \varphi_1(x) \leq y \leq \varphi_2(x)\},$$

for some continuous function  $\varphi_1, \varphi_2 : [a, b] \rightarrow \mathbb{R}, \varphi_1 \leq \varphi_2$ , or of the form

$$B' = \{(x, y) | c \leq y \leq d, \psi_1(y) \leq x \leq \psi_2(y)\},$$

for some continuous function  $\psi_1, \psi_2 : [c, d] \rightarrow \mathbb{R}, \psi_1 \leq \psi_2$ . We call these domains as **elementary domains**.

For the elementary domains, using Theorem 1.7, we get

$$\begin{aligned} P\{(X, Y) \in B\} &= \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx, \\ P\{(X, Y) \in B'\} &= \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy, \end{aligned}$$

where  $(X, Y)$  is with pdf  $f$ .

Now we will give some important class of multi valued distribution functions.

**Example 0.6** (*Uniform distribution on disc of radius  $R$* ) Consider the function

$$f(x, y) = \begin{cases} \frac{1}{\pi R^2} & \text{if } x^2 + y^2 \leq R^2 \\ 0 & \text{if otherwise.} \end{cases}$$

Then  $f$  is called joint density of uniform distribution on  $\{(x, y) | x^2 + y^2 \leq R^2\}$ . In fact student can try to construct a pair of random variables  $(X, Y)$  on some probability space  $(\Omega, \mathcal{F}, P)$  such that its joint distribution function is given by

$$P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(x, y) dx dy.$$

Any Guess! Hint: see how uniform  $(0, 1)$  random variable is constructed.

**Example 0.7** Let  $X, Y$  be two random variables with joint pdf given by

$$f(x, y) = \frac{\sqrt{3}}{4\pi} e^{-\frac{(x^2 - xy + y^2)}{2}}, \quad x, y \in \mathbb{R}.$$

If  $f_X(\cdot), f_Y(\cdot)$  denote the marginal pdfs of  $X$  and  $Y$  respectively, then

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= \frac{\sqrt{3}}{4\pi} \int_{-\infty}^{\infty} e^{-\frac{(x^2 - xy + y^2)}{2}} dy \\ &= \frac{\sqrt{3}}{4\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - \frac{x}{2})^2} e^{-\frac{3x^2}{8}} dy \\ &= \frac{\sqrt{3}}{4\pi} e^{-\frac{3x^2}{8}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(y - \frac{x}{2})^2} dy \\ &= \frac{\sqrt{3}}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{3x^2}{8}}. \end{aligned}$$

Therefore

$$X \sim N(0, \frac{4}{3}).$$

Here  $X \sim N(m, \sigma^2)$  means  $X$  is normally distributed with parameters  $m$  and  $\sigma^2$ . Similarly,

$$\begin{aligned} f_y(y) &= \int_{-\infty}^{\infty} f(x, y) dy \\ &= \frac{\sqrt{3}}{2} \frac{1}{\sqrt{2\pi}} e^{-\frac{3y^2}{8}}. \end{aligned}$$

Therefore

$$Y \sim N(0, \frac{4}{3}).$$

Also note that  $X$  and  $Y$  are dependent since,

$$f(x, y) \neq f_X(x)f_Y(y),$$

see exercise.

**Real valued transformations of multi random variables:** In this subsection, we look at transformations of  $(X, Y)$  of the form  $\varphi \circ (X, Y)$ , i.e.  $\varphi(X, Y)$  where  $\varphi : \mathbb{R}^2 \rightarrow \mathbb{R}$ , i.e. real valued functions of  $(X, Y)$ , for example  $X+Y, XY$  etc. One can write down the distribution function  $Z = \varphi \circ (X, Y)$ . Set

$$A_z = \{(x, y) | \varphi(x, y) \leq z\}, \quad z \in \mathbb{R}.$$

Then

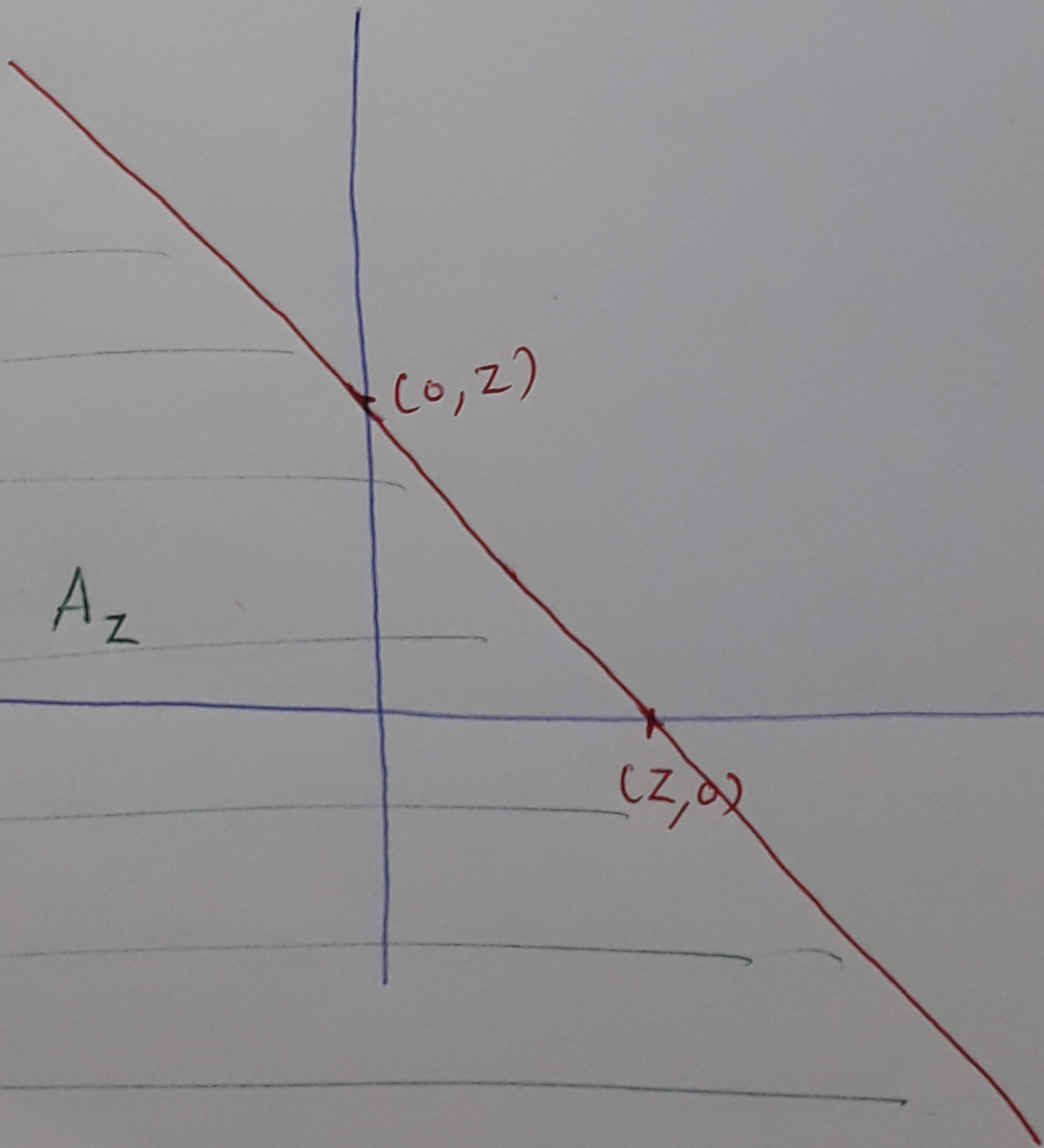
$$F_Z(z) = P(\{\varphi(X, Y) \in (-\infty, z]\}) = P((X, Y) \in A_z),$$

where  $F_Z$  denote the distribution function of  $Z = \varphi(X, Y)$ . Hence it is all about understanding  $A_z$  and then computing  $P((X, Y) \in A_z)$ .

**Example 0.8** (Distribution of sum) Let  $X, Y$  be with joint pdf  $f$ . Then one can compute the distribution of the sum  $Z = X + Y$  as follows. Note

$$A_z = \{(x, y) | -\infty < x < \infty, -\infty < y \leq z - x\}$$

is an elementary domain, in fact it is the region 'below' the line  $x + y = z$ .



Hence

$$\begin{aligned}
 F_Z(z) &= \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f(x, y) dy dx \\
 (\text{put } t = y + x) &= \int_{-\infty}^{\infty} \int_{-\infty}^z f(x, t - x) dt dx \\
 (\text{change order of integration}) &= \int_{-\infty}^z \left[ \int_{-\infty}^{\infty} f(x, t - x) dx \right] dt.
 \end{aligned}$$

Hence  $X + Y$  has pdf given by

$$f_Z(z) = \int_{-\infty}^{\infty} f(x, z - x) dx.$$

**Corollary 0.1** *Let  $X, Y$  be independent random variables with joint pdf  $f$ . Then the pdf of  $X + Y$  is given by*

$$f_{X+Y}(z) = f_X * f_Y(z),$$

where  $f_X * f_Y$  denote the convolution of  $f_X$  and  $f_Y$  and is defined as

$$f_X * f_Y(z) = \int_{\mathbb{R}} f_X(x) f_Y(z - x) dx = \int_{\mathbb{R}} f_X(z - y) f_Y(y) dy.$$

**Proof.** The proof (exercise) follows immediately from  $f(x, y) = f_X(x) f_Y(y)$  and the above example.